Constipation in Children

70

Introduction

Chronic constipation in children is a very common and increasingly encountered problem in office practice. While <5% children with constipation have a definitive organic etiology (e.g., congenital, surgical, neurological, and endocrine conditions) for constipation, the vast majority of children have no proven cause and are labeled as functional constipation. The normal stool frequency in children is mentioned in **Table 1**.

TABLE 1: Defecation frequency in children.

Age	*Average number of stools per day*
<1 month	3–4
1 month to 1 year	1.5–2
1–2 years	1–2
2–18 years	1

Recognition of constipation is not based on stool frequency alone and the physician should assess other parameters like stool consistency/associated pain or struggle during defecation/associated fissure/soiling of underwear before labeling it as constipation.

Definition

Functional constipation is considered when any of the *two or more criteria* to be fulfilled to label as constipation. The criteria include:

- ☑ Stool frequency ≤2 per week
- ☑ History of hard and painful bowel movements
- ☑ History of retention of stools
- ☑ Large diameter stool obstructing the Indian pots
- ☑ Fecal incontinence
- ☑ Per abdomen or per rectum examination reveals fecal mass.

Clinical Evaluation

- ☑ The history taking and examination must include information regarding time of onset of constipation, passage of meconium, growth and developmental milestones, and history of withholding in order to differentiate functional constipation from organic causes **(Table 2)**.
- ☑ The presence of a mass or palpable fecaloma on abdominal examination might suggest the presence of fecal impaction (inability to pass/evacuate hard and large diameter stools) which often causes abdominal pain or fecal soiling.
- ☑ A perianal examination (for the presence of fissure/sentinel skin tags) and examination of the spinal area for any possible underlying spinal/neurological conditions is recommended.

The history taking should include the following points in detail:

- ☑ *Age of onset of symptoms*: Surgical causes are more common in infantile onset constipation. Functional constipation typically occurs later.
- ☑ *Passage of meconium after birth*: Delayed passage of meconium is observed with Hirschsprung's disease.
- ☑ *Stool characteristics*: Amount, consistency, and frequency of stools.
- ☑ Pain during defecation and blood streaks on stool or drops after stools may suggest presence of anal fissure.
- ☑ Retentive postures during defecation are often a "habit" mostly seen in younger children with functional constipation who dread the act of passing painful stools.
- ☑ *Urinary symptoms*: Increased frequency, burning micturition, and urinary incontinence may suggest coexisting urinary tract infection or voiding dysfunction.

TABLE 2: Red flag symptoms and signs pointing to common organic causes.

Signs and symptoms	***Probable cause for constipation***
Short stature, decrease in height velocity, delayed development/poor scholastics, lethargy, cold intolerance in older children	Hypothyroidism
Delayed passage or failure to pass meconium, failure to thrive, abdominal distension, history of recurrent enterocolitis	Hirschsprung's disease
Tuft of hair at back of the spine or any other abnormal neurological signs in lower limbs	Spinal cord abnormalities and spinal dysraphism
Significant perinatal history, delayed development, feeding abnormalities, regression of acquired milestones	Cerebral palsy and neurodegenerative disorder
Recurrent respiratory infections, meconium ileus, and failure to thrive	Cystic fibrosis

Bristol Stool Chart

The Bristol stool chart is a helpful handy chart for assessing the type of stools in clinical practice **(Fig. 1)**. It can be kept in the consultation room.

Type 1	Separate hard lumps, like nuts (hard to pass)
Type 2	Sausage-shaped but lumpy
Type 3	Like a sausage with cracks on its surface
Type 4	Like as sausage, smooth and soft
Type 5	Soft blobs, clear cut edges (passed easily)
Type 6	Fluffy pieces, ragged edges mushy stool
Type 7	Waterly, no solid pieces, entirely liquid

Fig. 1: Bristol stool chart.

(*Source*: Lewis SJ, Heaton KW. Stool form scale as a useful guide to intestinal transit time. Scand J Gastroenterol. 1997;32(9):920-4.)

Management

The diagnosis of functional constipation can be made with a good clinical history and physical examination and seldom needs many investigations. The presence of red flag signs often point toward an organic etiology and such children should be appropriately investigated.

The management of constipation involves five components and is depicted in the **Figure 2**.

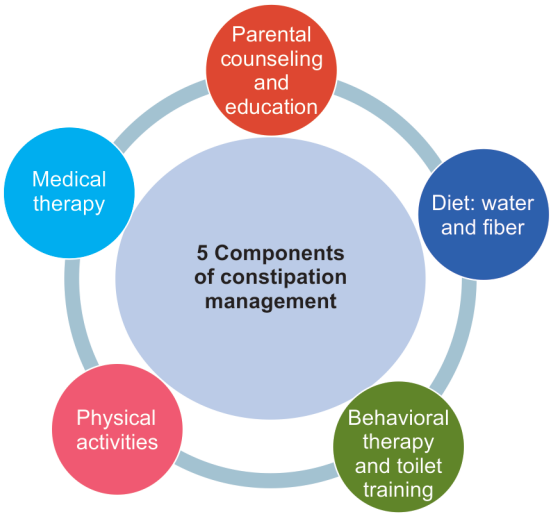

Fig. 2: Five components in the management of functional constipation.

Management

1. **Parental counseling and education:**
   - Parents need to be educated regarding the pathophysiology of functional constipation so that they understand why their child struggles to defecate and what needs to be done to improve the situation.
   - The longer the stool is retained in the colon, the drier/harder and bulkier it becomes and this causes painful evacuation. The withholding behavior of a child often begins after such a painful movement of the bowel and when this continues for a period of time; it establishes a vicious cycle that leads to chronic habitual constipation.
   - This can lead to fecal impaction/overflow incontinence when left untreated. Long-standing constipation (left untreated for several months) in a small percentage of children can lead to the development of a secondary megarectum.
   - These children require specialist attention, evaluation, and several months of effective laxative therapy before the megarectum and constipation resolve.

   Emphasis also should be given to the need for compliance and regular medical follow-up until complete recovery. Parental anxiety regarding the side effects of long-term osmotic laxatives should be allayed.
2. **Diet, fiber, and water intake:**
   - The daily diet should include a sufficient quantity of fiber (0.5 g/kg/day) and adequate water. It should be noted that milk has a minimal quantity of fiber. The daily diet should include cereals, pulses, vegetables, and fruits.
   - Milk however may be allowed in permissible quantities as per child's age. It is important to provide symptom relief to the child with medications and not rely on diet alone to treat a child with chronic constipation.
3. **Physical activity:** Sedentary lifestyle is discouraged and participation in physical activities are encouraged as this encourages bowel movement.
4. **Toilet training:** The parents are advised to encourage their child to defecate within 30 minutes of the major meal in order to utilize gastrocolic reflex. If the child uses Western toilet, the child should be encouraged to use a footrest to make an effective angulation between abdomen and thighs to facilitate proper passage of stools. (The Indian style closet is considered anatomically more ideal for defecation than the Western style closet).
5. **Medical therapy:** Medical therapy includes disimpaction and maintenance.

Identification and Treatment of Fecal Impaction

A significant proportion of children with long-standing constipation often have very hard and impacted stools, which may not be cleared with regular/maintenance dose laxatives. Identification of fecal impaction and disimpaction with laxatives is an important initial requisite for subsequent effective maintenance laxative therapy. Different regimes are mentioned in **Table 3**. Oral agents are preferred over rectal agents.

TABLE 3: Different regimens used for fecal disimpaction.

Home-based disimpaction regimen	***Dosage***	***Side effects/comments***
Polyethylene glycol	1.5–2 g/kg/day in two divided doses for 3–6 days	Well tolerated but slower to act
Oral agent—Polyethylene glycol	25 mL/kg/h orally or by nasogastric tube over 4–6 hours	☑ Requires admission ☑ Ensures disimpaction of the entire colon within a few hours ☑ May cause bloating, abdominal distension, and nausea
Rectal agent, e.g., sodium phosphate enemas	2.5 mL/kg (2–18 years old)	

Maintenance Therapy

- ☑ The two classes of laxatives used in the maintenance phase are osmotic laxative and stimulant laxatives **(Table 4)**.
- ☑ Osmotic laxatives are the preferred first choice of the pediatric medical fraternity due to: (1) extensive experience and published literature and (2) better side effect profile.
- ☑ Successful treatment is defined as regular painless defecation, achievement of toilet training, and absence of fecal soiling or blood in stools. Stool diary is a useful monitoring tool that often ensures better compliance.
- ☑ The tapering of laxative dose is usually started after 2–3 months of successful treatment and continued over a period of 2–3 months.
- ☑ Avoid abrupt stopping of the laxatives. Stimulant laxatives such as senna and bisacodyl are often used as a *rescue therapy* for 2–3 days whenever children suffer from relapse of acute constipation while on *maintenance therapy*.

Management

Maintenance Therapy

The children who do not respond to the sustained optimal treatment should be referred for expert opinion.

TABLE 4: Commonly used laxatives and dose used in children and their mechanism of action.

Types of laxatives	*Name of laxative*	*Dose*	*Mechanism of action*	*Side effect*
Osmotic laxatives	Polyethylene glycol	0.5–1 g/kg/day	Large inert molecule which retains intraluminal water	Nausea and bloating
	Lactulose	<1 year 2.5 mL BD 1–<5 years: 2.5–10 mL BD 5–20 years 5–20 mL BD	Nonabsorbable synthetic disaccharides product which retains intraluminal water	Bloating and abdominal discomfort
	Lactitol	200–400 mg/kg/day	Synthetic monohydrate product which retains intraluminal water	Mild bloating
Stimulant laxatives	Sodium picosulfate	2.5–10 mg per day up to 4 years 2.5–20 mg per day up from 4–18 years	Converts into active metabolite by intestinal bacteria which increases the peristalsis	Nausea, crampy abdominal pain, occasionally loose motions

Points to Remember

- ☑ The diagnosis of functional constipation may be considered in the absence of red flag signs.
- ☑ Organic causes are more common in infantile onset constipation.
- ☑ Identification and treatment of fecal impaction is important for successful *maintenance* laxative therapy.
- ☑ Long-term maintenance laxative therapy along with toilet training is often needed for best outcome.
- ☑ Education and counseling of parents is important to ensure compliance and successful outcomes.
- ☑ Refractory constipation should be referred to a pediatric gastroenterologist for further evaluation.

Further Reading

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- ☑ Tabbers MM, DiLorenzo C, Berger MY, Faure C, Langendam MW, Nurko S, et al. Evaluation and treatment of functional constipation in infants and children: evidence-based recommendations from ESPGHAN and NASPGHAN. J Pediatr Gastroenterol Nutr. 2014;58(2):258-74.
- ☑ Yachha SK, Srivastava A, Mohan N, Bharadia L, Sarma MS, Indian Society of Pediatric Gastroenterology, et al. Management of Childhood Functional Constipation: Consensus Practice Guidelines of Indian Society of Pediatric Gastroenterology, Hepatology and Nutrition and Pediatric Gastroenterology Chapter of Indian Academy of Pediatrics. Indian Pediatr. 2018;55(10):885-92.

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